HITEC University Taxila

Department of Cybersecurity

CS-304 Database Systems

Complex Computing Problem

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1. Introduction

The Employee Management System (EMS) is designed to automate and streamline employee management tasks in an organization. The system allows administrators to manage departments, designations, employees, salaries, attendance, and leave requests efficiently through a user-friendly web portal.

Key features include:

- **CRUD operations** for all entities (Create, Read, Update, Delete)
- **Relational database** to maintain data integrity
- **User login system** for secure access
- **Bootstrap GUI** for easy navigation and responsive design
- **Data validation and foreign key constraints** to prevent errors

2. Objectives

The main objectives of this project are:

1. To provide a centralized system to manage employee data
2. To automate repetitive HR tasks like attendance tracking and leave management
3. To maintain accurate records of salaries and employee details
4. To provide a secure login system for administrators
5. To demonstrate practical application of PHP, MySQL, and web development concepts

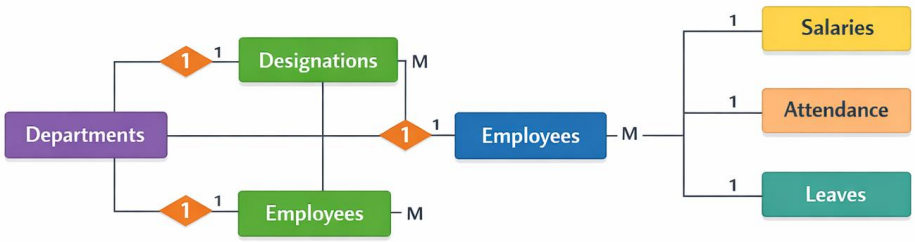
3. Database Design

The system uses a **MySQL relational database** with **7 tables**:

Table Name	Purpose
Departments	Stores department details like name and location
Designations	Stores designations with department reference
Employees	Stores employee personal and professional information
Salaries	Stores salary details of employees
Attendance	Tracks daily attendance of employees
Leaves	Manages leave requests and approvals
Users	Stores login credentials of administrators

3.1 Entity Relationship Diagram (ERD)

```
Departments (1) --- (M) Designations
Departments (1) --- (M) Employees
Designations (1) --- (M) Employees
Employees (1) --- (M) Salaries
Employees (1) --- (M) Attendance
Employees (1) --- (M) Leaves
```



4. Database Normalization

The database is normalized up to **3NF**:

1. **1NF (First Normal Form):**

    - All tables have unique primary keys
    - No repeating groups or arrays
2. **2NF (Second Normal Form):**
    - All non-key attributes fully depend on primary key
    - Example: `Employees` table → email, phone depend on employee_id
3. **3NF (Third Normal Form):**
    - No transitive dependencies
    - Example: `Designations` table → title depends only on designation_id, not department_name

5. Portal Development

Technology Used:

- PHP (Server-side scripting)
- MySQL (Database)
- Bootstrap 5 (Frontend UI)

5.1 Features:

1. **User Authentication:**
    - Secure login page
    - Only authenticated users can access the dashboard
2. **Department Management:**
    - Add new departments with name and location
    - Data stored in `Departments` table
3. **Designation Management:**
    - Add designations linked to departments
    - Prevents orphaned records through foreign key constraints
4. **Employee Management:**
    - Add, update, and delete employees
    - Assign department and designation
    - Store personal information (name, email, phone, hire date)
5. **Salary Management:**
    - Add employee salary details including allowances and deductions
6. **Attendance Tracking:**
    - Record daily attendance (Present/Absent) for employees
7. **Leave Management:**
    - Add leave requests with start date, end date, type, and status
    - Leave records linked to employees
8. **View Employees:**
    - View consolidated employee data with JOINs across tables
    - Includes department, designation, salary, attendance, leave status

5.2 User Interface

- **Navigation:** Bootstrap pills for easy navigation between modules
- **Forms:** Bootstrap cards for each module with validation
- **Tables:** Striped and bordered tables for data display
- **Feedback:** Success and error messages after operations

6. CRUD Operations

The system implements full **CRUD functionality**:

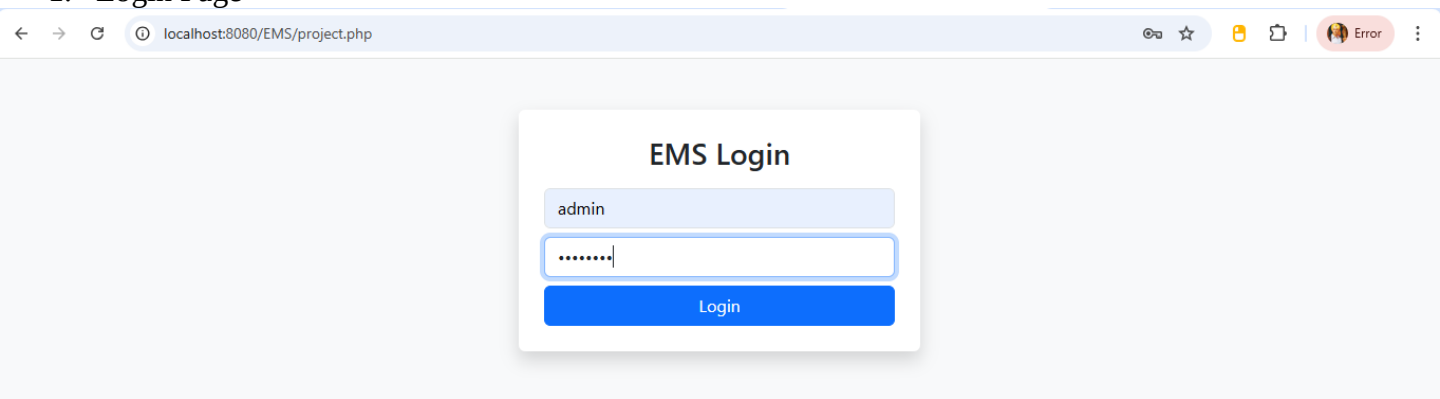
Operation	Description
Create	Add departments, designations, employees, salaries, attendance, leave
Read	View employee data with combined information using JOIN queries
Update	Modify existing employee, salary, or leave records
Delete	Remove employees or related records safely using foreign key constraints

Error Handling:

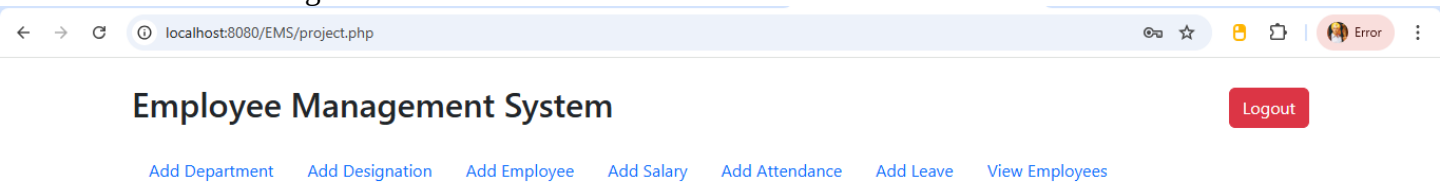
- Prepared statements prevent SQL injection
- Required fields are validated
- Foreign key constraints prevent orphaned records

7. Sample Screenshots

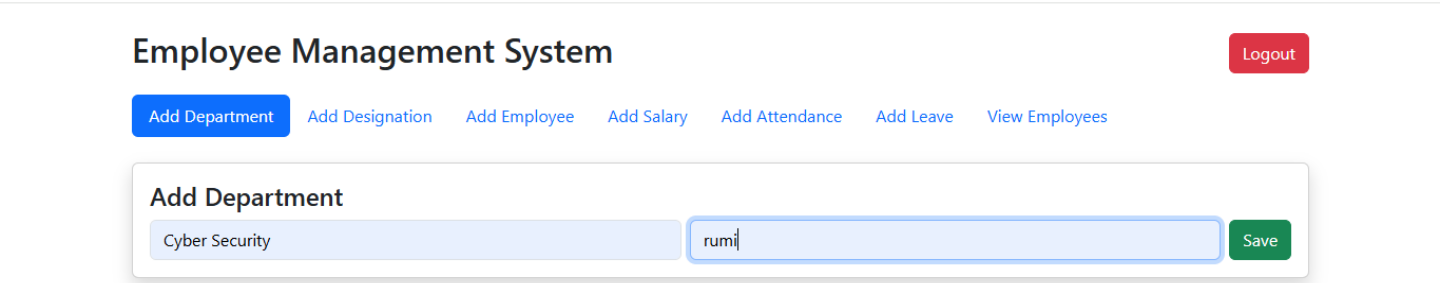
1. Login Page



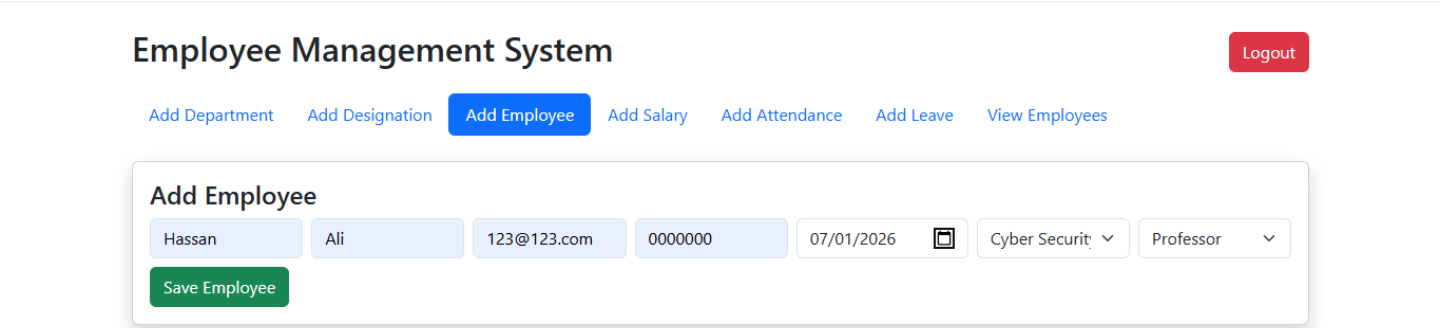
2. Dashboard Navigation



3. Add Department Form



4. Add Employee Form



5. View Employees Table

8. Challenges Faced

1. Handling foreign key constraints while inserting data
2. Ensuring all forms validate inputs properly
3. Creating a single-file PHP project while keeping it maintainable
4. Implementing Bootstrap for all forms and tables

9. Conclusion

The Employee Management System successfully demonstrates:

- Database design and normalization
- Full CRUD operations on multiple tables
- A functional web portal using PHP and MySQL
- Secure login system and proper error handling

This project can be extended further to include:

- Role-based access control
- Salary calculations and payslips
- Reports and charts for HR analytics